

Functional Annotation of LolD (Q88KY4 / PP_2155) in *Pseudomonas putida* KT2440

0. Identity verification (performed first)

- **UniProt:** Q88KY4 — "Lipoprotein-releasing system ATP-binding protein LolD"; EC 7.6.2.- (translocase / ABC-transporter ATPase).
- **Gene / locus:** *lolD* / PP_2155, *Pseudomonas putida* KT2440.
- **Family / domains:** ABC transporter superfamily (LolD subfamily); AAA+ ATPase (IPR003593), ABC transporter-like ATP-binding domain (IPR003439/IPR017871), LolD-like ATP-binding (IPR015854), and the diagnostic **LolD lipoprotein-release ATP-binding** signature (IPR011924).

Verification result: CONFIRMED, unambiguous. The gene symbol *lolD*, the "lipoprotein-releasing system ATP-binding protein" description, the EC 7.6.2.- translocase class, and the LolD-specific InterPro signature (IPR011924) all coincide with a single, well-defined function: the ATPase subunit of the Lol (Localization of lipoprotein) ABC transporter LolCDE. The symbol is **not** ambiguous — "LolD" is used consistently across Gram-negative bacteria for this exact subunit. There is no competing gene family sharing the symbol. No *P. putida*-specific primary study of PP_2155 exists, so the functional narrative below is built from (i) the extensively characterized *Escherichia coli* system and (ii) the experimentally validated *Pseudomonas aeruginosa* orthologous system, then transferred to *P. putida* by orthology.

1. Summary (answer to the research question)

LolD (Q88KY4, PP_2155) is the **cytoplasmic ATP-binding/ATP-hydrolyzing (nucleotide-binding-domain) subunit of the LolCDE ABC transporter**, the machine that initiates sorting of outer-membrane lipoproteins in Gram-negative bacteria. Two copies of LolD associate with one integral-membrane LolC and one LolE to form LolCDE in the **inner (cytoplasmic) membrane**; LolD hydrolyzes ATP on the **cytoplasmic face** of that membrane, and this chemical energy is mechanically coupled — via the conserved "LolD motif" that contacts LolC/LolE — to the **detachment (release) of triacylated outer-membrane lipoproteins from the**

periplasmic leaflet of the inner membrane. Unusually for an ABC transporter, LolCDE does not translocate a solute across the bilayer; instead LolD's ATPase activity powers extraction of the lipoprotein's lipid anchor and its hand-off to the periplasmic chaperone LolA, the committed first step of the Lol trafficking pathway. The reaction is essential for outer-membrane biogenesis and viability, making LolCDE (and thus LolD) a validated antibacterial target.

Confidence: Very high. The annotation rests on four independent, converging evidence layers: (1) direct *E. coli*/*P. aeruginosa* biochemistry establishing LolD's reaction, mechanism and essentiality; (2) *P. putida*-specific sequence analysis showing a complete, canonical ABC nucleotide-binding domain (NBD-only architecture); (3) 81.5% identity to the biochemically validated *P. aeruginosa* LolD ortholog; and (4) an intact chromosomal *lolC-lolD-lolE* cluster in *P. putida* KT2440, confirming the full machine is encoded in the target organism. No experimental study of PP_2155 itself exists, but function transfer is exceptionally well supported.

2. Biological context: the lipoprotein biogenesis and Lol sorting pathway

Bacterial lipoproteins are made as cytoplasmic precursors, exported by Sec to the periplasmic (outer) leaflet of the inner membrane, and matured by three enzymes — Lgt (diacylglycerol transferase), LspA (signal peptidase II), and Lnt (apolipoprotein N-acyltransferase) — to yield the N-terminal triacyl-cysteine lipid anchor (Narita & Tokuda 2010,).

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E. coli has >90 lipoproteins, most destined for the outer membrane (Narita & Tokuda 2010; Narita, Matsuyama & Tokuda 2004,).

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Sorting of the outer-membrane subset requires the **five-protein Lol system, LolABCDE** (Okuda & Tokuda 2011,):

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1. **LolCDE** (inner-membrane ABC transporter; contains **LolD**) recognizes an outer-membrane-destined lipoprotein and, using ATP, **releases it from the inner membrane**, generating a soluble LolA–lipoprotein complex.
2. **LolA** (periplasmic chaperone) shields the acyl chains in a hydrophobic cavity and ferries the lipoprotein across the aqueous periplasm.
3. **LolB** (outer-membrane receptor lipoprotein) accepts the cargo and **inserts it into the inner leaflet of the outer membrane** (Narita & Tokuda 2010,).

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LolD is the energizing engine of step 1 — the committed, rate-initiating step of the whole pathway.

3. Primary molecular function of LolD

3.1 It is the ATPase subunit of LolCDE

The founding study reconstituted LolCDE in proteoliposomes and showed it "belong[s] to the ATP-binding cassette (ABC) transporter family" and "catalyses the release of lipoproteins in LolA- and sorting-signal-dependent manners" (Yakushi et al. 2000,).

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Notably, the authors emphasized that "the LolCDE complex differs mechanistically from all other ABC transporters as it is not involved in the transmembrane transport of substrates" — i.e., LolD's ATP hydrolysis powers a *release/extraction* reaction, not vectorial solute transport.

3.2 Stoichiometry and site of catalysis

LolCDE contains "two copies of an ATPase subunit, LolD, and one copy each of integral membrane subunits LolC and LolE. **LolD hydrolyzes ATP on the cytoplasmic side of the inner membrane**, while LolC and/or LolE recognize and release lipoproteins anchored to the periplasmic leaflet of the inner membrane" (Ito et al. 2006,

).

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Thus the catalytic reaction — $\text{ATP} + \text{H}_2\text{O} \rightarrow \text{ADP} + \text{P}_i$ — occurs in the **cytoplasm**, at the two LolD nucleotide-binding domains, while the substrate-handling occurs on the opposite (periplasmic) face.

3.3 Substrate specificity

LolCDE's substrate is the set of **mature triacylated lipoproteins bearing an outer-membrane sorting signal**. Specificity is governed by the residue(s) following the N-terminal acyl-Cys: in *E. coli* the "+2 rule" (Asp at position 2 = inner-membrane retention / LolCDE-avoidance signal), and in *Pseudomonas* the "+3,+4 rule" (Narita & Tokuda 2007,

).

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LolCDE releases only lipoproteins **lacking** the retention signal. LolD itself does not read the sorting signal (that is done by LolC/LolE); LolD supplies the energy that acts on whatever cargo the membrane subunits have engaged. Aminoacylation of the N-terminal cysteine (Lnt product) is required for Lol-dependent release (Fukuda et al. 2002,

).

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3.4 Mechanistic coupling — the "LolD motif"

LolD carries "a characteristic sequence called the LolD motif, which is highly conserved among LolD homologs but not other ABC transporters of *E. coli*" (Ito et al. 2006,

).

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Systematic mutagenesis of all 32 residues of this motif produced 26 dominant-negative alleles (growth arrest on overexpression), and suppressor mutations mapped to the periplasmic loops of LolC and cytoplasmic loops of LolE — genetically defining the **LolD↔LolC/E interaction surface** through which ATP hydrolysis is transmitted to the membrane subunits to drive release. This is the mechanistic heart of energy coupling: LolD is the motor, LolC/E the substrate-handling stators.

3.5 Biochemical reconstitution

The functional machine was rebuilt from separately purified subunits: "the functional lipoprotein-releasing machinery was reconstituted into proteoliposomes with *E. coli* phospholipids and separately purified LolC, LolD and LolE" (Kanamaru et al. 2007,

).

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LolD's ATPase activity (and its dependence on phospholipids/assembly) was directly measured, and — strikingly — a minimal releasing activity could form from LolE + LolD without LolC, underscoring LolD as the indispensable catalytic partner.

4. Subcellular localization / where LolD acts

- **Compartment:** peripheral membrane protein of the **inner (cytoplasmic) membrane**, associated with the integral-membrane subunits LolC/LolE.
- **Catalytic site orientation:** the two LolD nucleotide-binding domains face the **cytoplasm**, where ATP hydrolysis occurs (Ito et al. 2006,

).

P 16585747

- **Site of the biological reaction it powers:** extraction of lipoproteins from the **periplasmic leaflet of the inner membrane**, with product delivery into the **periplasm** (as the LolA–lipoprotein complex). Downstream steps (LolA transit, LolB insertion) occur in the periplasm and outer membrane respectively (Okuda & Tokuda 2011,

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5. Transfer to *Pseudomonas putida* (orthology evidence)

No PP_2155-specific experimental study exists. Confidence in the annotation rests on:

- **Close-relative validation:** In *P. aeruginosa*, cloned LolCDE (with its LolD ATPase), LolA and LolB homologs are functional — purified LolCDE "reconstituted into proteoliposomes ... When incubated in the presence of ATP and a LolA homologue, the reconstituted LolCDE homologue released lipoproteins, leading to the formation of a LolA-lipoprotein complex," with subsequent LolB-dependent outer-membrane insertion (Tanaka et al. 2007,

).

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P. putida is a close congener of *P. aeruginosa*.

- **Pseudomonas sorting logic:** the "+3,+4 rule" defines the inner-membrane retention signal read by the *Pseudomonas* Lol system (Narita & Tokuda 2007,

P 17350956

— the specificity framework applicable to *P. putida*.

- **Sequence/domain conservation:** Q88KY4 bears the LolD-diagnostic InterPro signature (IPR011924) plus the canonical Walker A/B and ABC signature motifs (IPR003439/ IPR017871), i.e., a bona fide LolD-type nucleotide-binding domain.

5.1 Organism-specific sequence evidence (this work)

Direct motif analysis of the retrieved 227-residue Q88KY4 sequence confirms a **complete, canonical ABC nucleotide-binding domain** in correct order and spacing:

Motif	Sequence (residues)	Role
Walker A / P-loop	GSSGSGKST (~42–50)	binds ATP β/γ phosphates
Q-loop	Q84 (in QFHH)	coordinates Mg^{2+} /catalytic water; couples to TMD partner
ABC signature / C-loop	LSGGERQR (~147–154)	forms composite ATP site at NBD dimer interface
Walker B	LVMLDE (~167–172; D171/E172)	Mg^{2+} coordination and catalytic base
H-loop / switch	H in VVTHD (~205–208)	stabilizes transition state

The protein is **only ~227 aa with no transmembrane segment** — i.e., a **stand-alone nucleotide-binding domain**, exactly the architecture expected for the separate cytoplasmic LolD ATPase that dimerizes and docks onto the integral-membrane LolC/LolE subunits. This constitutes direct *P. putida*-specific molecular evidence that Q88KY4 is a catalytically competent ABC ATPase of the LolD type, complementing the *E. coli*/*P. aeruginosa* biochemistry.

5.2 Quantitative orthology to experimentally validated LolD (this work)

Global (Needleman–Wunsch) pairwise alignment of the retrieved UniProt sequences:

Comparison	Identity	Notes
<i>P. putida</i> LolD (Q88KY4, 227 aa) vs <i>P. aeruginosa</i> LolD (Q9HZL7 / LOLD_PSEAE, 227 aa)	81.5% (185/227)	same length; <i>P. aeruginosa</i> ortholog is experimentally validated (P 17350955)
<i>P. putida</i> LolD vs <i>E. coli</i> LolD (P75957 / LOLD_ECOLI, 233 aa)	57.3% (134/234)	cross-genus; well above the ~30% "safe" ortholog-transfer threshold

All three proteins carry the **identical Walker A P-loop (GSSGSGKS)** and **identical ABC signature (LSGGE)** and are all annotated "Lipoprotein-releasing system ATP-binding protein LolD." The 81.5% identity to a biochemically validated same-genus ortholog, with perfectly conserved catalytic motifs and identical NBD-only architecture, elevates the annotation of PP_2155 from *inferred* to *strongly supported by quantitative homology to an experimentally characterized ortholog*.

5.3 Genomic context: a complete lolCDE cluster in *P. putida* (this work)

The loci flanking *lolD* in the *P. putida* KT2440 chromosome encode its two obligate membrane partners, in the canonical arrangement:

Locus	Gene	UniProt	Length	Product / localization
PP_2154	lolC	Q88KY5	416 aa	LolC; cell membrane, multi-pass
PP_2155	lolD	Q88KY4	227 aa	LolD ATPase; cell inner membrane, peripheral membrane protein
PP_2156	lolE	Q88KY3	413 aa	LolE "lipoprotein releasing system, permease protein"; cell membrane, multi-pass

Flanking genes (PP_2153, a PilZ-domain protein; PP_2157, a sensor histidine kinase) are unrelated. This chromosomal **lolC–lolD–lolE cluster** mirrors the *E. coli* *lolCDE* operon and demonstrates that *P. putida* encodes a **complete LolCDE transporter** with LolD's integral-membrane partners present and adjacently encoded — independent genomic-synteny evidence that PP_2155 is the ATPase subunit of an assembled LolCDE machine. The UniProt subcellular annotation for Q88KY4 — "Cell inner membrane; Peripheral membrane protein" (HAMAP-Rule MF_01708) — confirms the localization stated in §4.

6. Essentiality and pathway significance

The Lol pathway is required for viability; a phenotypic screen identified a pyrazole inhibitor whose resistance mutations map to "LolC or LolE, components of the **essential LolCDE transporter complex**, which is required for trafficking of lipoproteins to the outer membrane," and the compound blocked ATP-dependent lipoprotein release in spheroplasts (Nayar et al. 2015,

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Because LolD supplies the ATP hydrolysis that powers this essential release reaction, its activity is indispensable for outer-membrane biogenesis. This is the basis for interest in LolCDE/LolD as an antibacterial target.

7. Supported and refuted hypotheses

Supported - H1: Q88KY4 is the ATPase (NBD) subunit of the LolCDE ABC transporter — supported

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- H2: LolD hydrolyzes ATP in the cytoplasm and this energizes lipoprotein release from the inner-membrane periplasmic leaflet — supported

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- H3: Energy coupling proceeds via the conserved LolD motif contacting LolC/LolE — supported by mutagenesis/suppressor genetics

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- H4: Function is conserved in *Pseudomonas*, licensing orthologous annotation of PP_2155 — supported

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 - H5: The complex is essential and drug-targetable — supported
 (25733621).

Refuted / excluded - LolCDE is **not** a classical solute-translocating ABC transporter; it does not move substrate across the bilayer (explicitly stated in

10783239).
 LolD therefore is not annotated as energizing transmembrane import/export of small molecules. - The symbol *lolD* is **not** ambiguous with any unrelated gene family; no alternative identity was found.

8. Limitations and future directions

- **Organism gap:** All mechanistic data derive from *E. coli* and *P. aeruginosa*; direct *P. putida* KT2440 biochemistry/genetics on PP_2155 is lacking. A conditional depletion or purified-complex assay in *P. putida* would confirm the transfer.
- **Structure:** Cryo-EM structures of *E. coli* LolCDE exist (referenced in Tao et al. 2024,

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 and show a central lipoprotein-binding cavity; an experimental or AlphaFold model of *P. putida* LolD could confirm Walker A/B, signature, Q-loop and the LolD-motif interface, and quantify conservation.

- **Sorting signal:** Whether *P. putida* strictly follows the +3,+4 rule for its own lipoprotein repertoire has not been individually tested.

References (PMIDs)

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